Chapter 20

The Second Law of Thermodynamics

Learning Outcomes

In this chapter, you'll learn…

- the difference between reversible and irreversible processes.
- the physics of internal-combustion engines.
- how refrigerators and heat engines are related, and how to analyze the performance of a refrigerator.
- how the second law of thermodynamics sets limits on the efficiency of engines and the performance of refrigerators.
- what is meant by entropy, and how to use this concept to analyze thermodynamic processes.

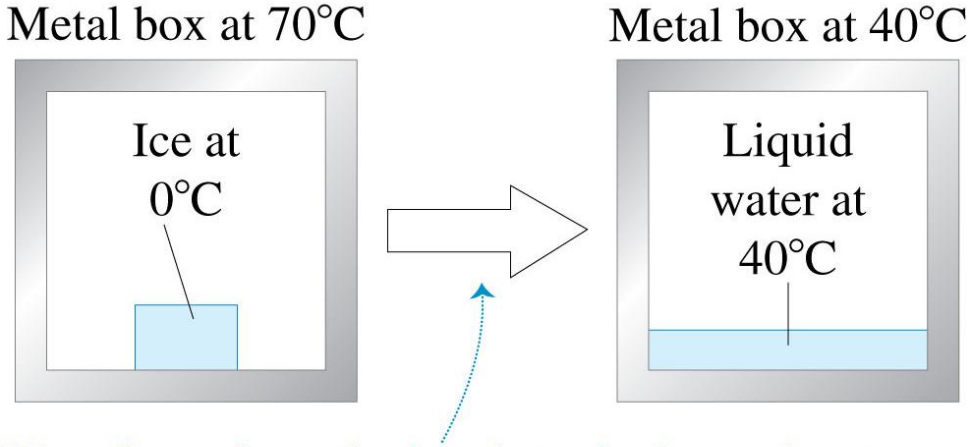
Introduction

- Why does heat flow from the hot lava into the cooler water?
- Could it flow the other way?

- It is easy to convert mechanical energy completely into heat, but not the reverse. Why not?
- We need to use the second law of thermodynamics and the concept of entropy to answer the above questions.

Directions of Thermodynamic Processes (1 of 2)

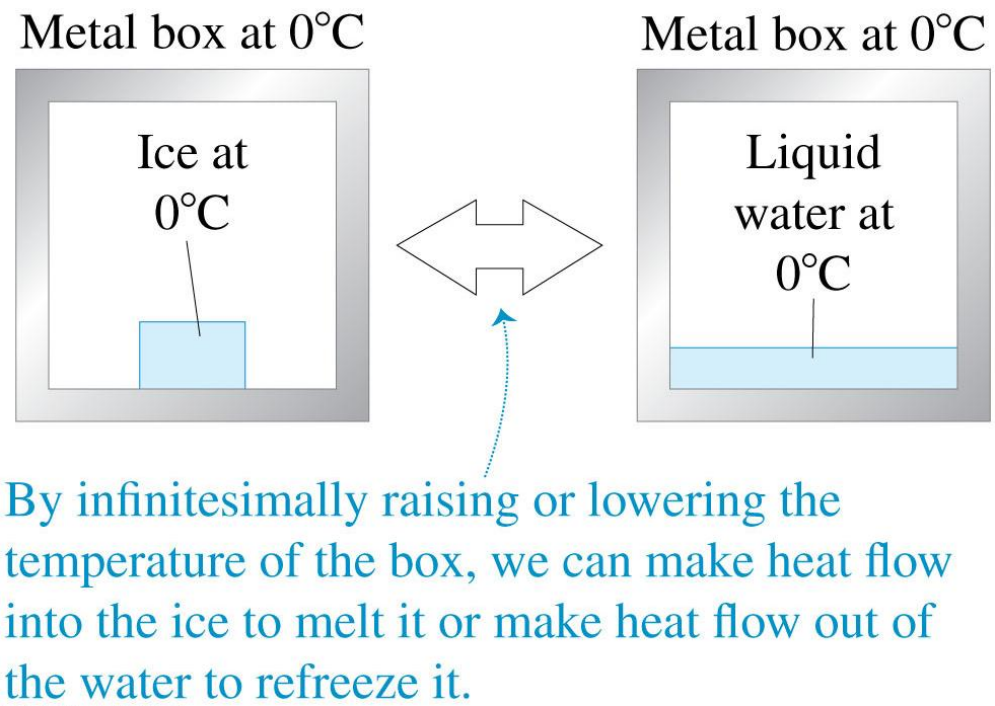
- The direction of a **reversible process** can be reversed by an infinitesimal change in its conditions.
- The system is always in or very close to thermal equilibrium.
- For example, a block of ice melts **irreversibly** when we place it in a hot metal box.



Heat flows from the box into the ice and water, never the reverse.

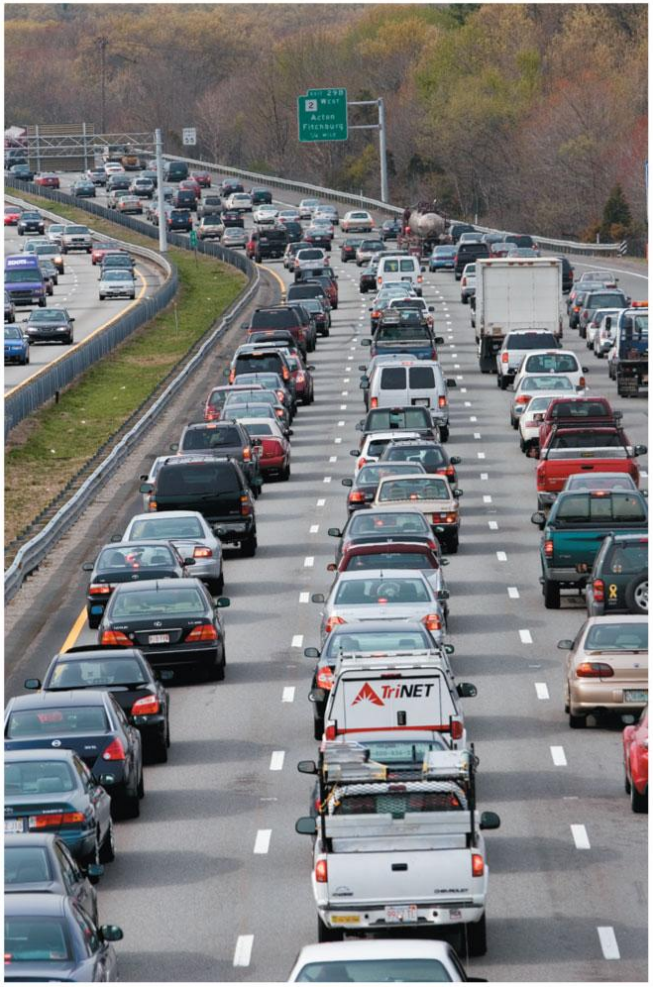
Directions of Thermodynamic Processes (2 of 2)

- A block of ice at 0 degrees Celsius can be melted **reversibly** if we put it in a 0 degrees Celsius metal box.



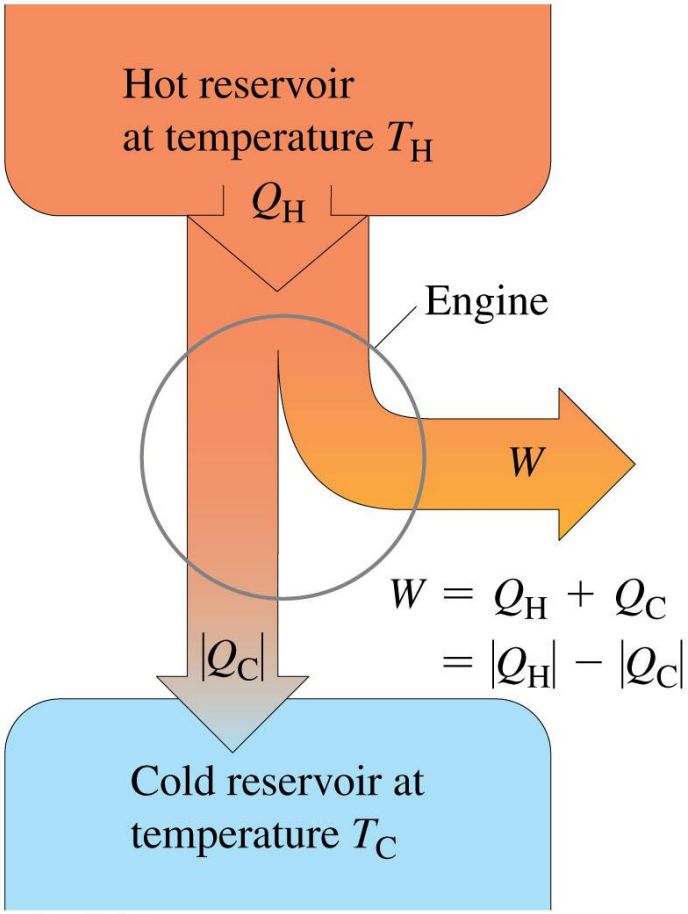
Heat Engines (1 of 2)

- A **heat engine** is any device that partly transforms heat into work or mechanical energy.
- All motorized vehicles other than purely electric vehicles use heat engines for propulsion.
- (Hybrid vehicles use their internal-combustion engine to help charge the batteries for the electric motor.)



Heat Engines (2 of 2)

- Simple heat engines operate on a **cyclic process** during which they absorb heat Q_H from a hot reservoir and discard some heat Q_C to a cold reservoir.
- Shown is a schematic energy-flow diagram for a heat engine.



The Efficiency of a Heat Engine

- The **thermal efficiency *e*** of a heat engine is the fraction of Q_H that is converted to work.

Work done by engine Heat rejected by engine

Thermal efficiency of an engine

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left|\frac{Q_C}{Q_H}\right|$$

Heat absorbed by engine

- *e* is what you get divided by what you pay for.
- This is always less than unity, an all-too-familiar experience!

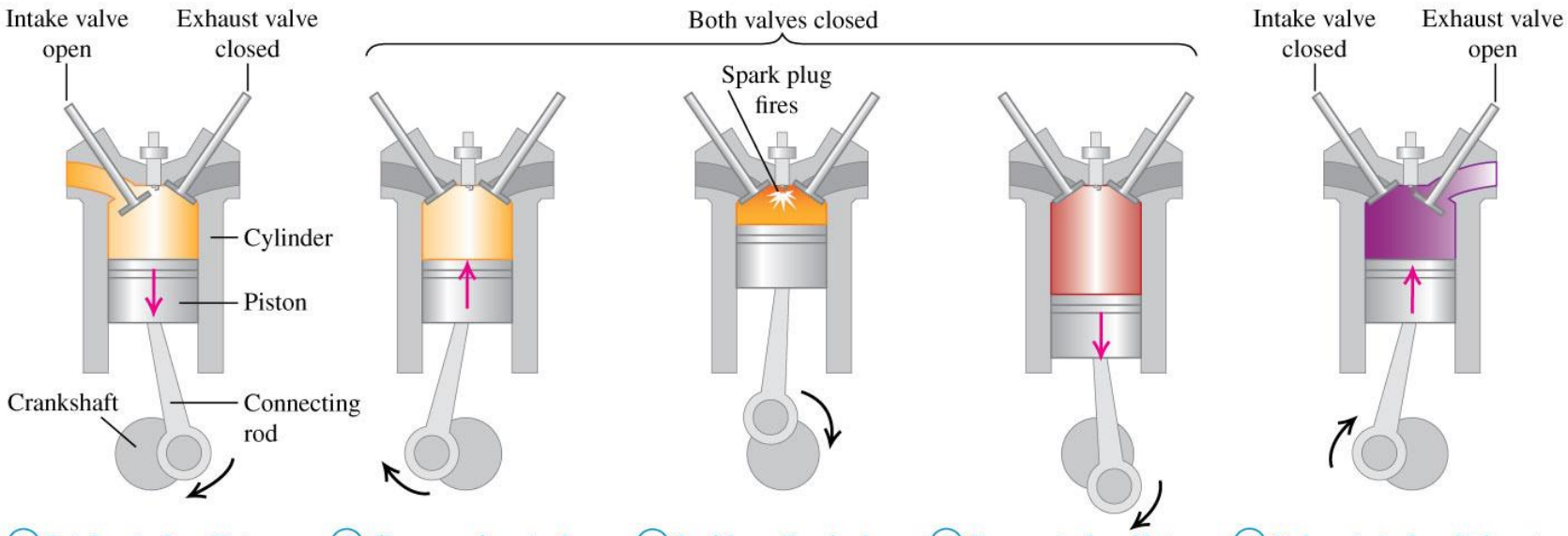
Question 1

10 pts

An engine with an efficiency 0.29 performs 1,049 J of work during a typical cycle. How much energy does it exhaust to the cold reservoir in one cycle?

2,568.2414

Internal-Combustion Engines



(1) **Intake stroke:** Piston moves down, causing a partial vacuum in cylinder; gasoline–air mixture enters through intake valve.

(2) **Compression stroke:** Intake valve closes; mixture is compressed as piston moves up.

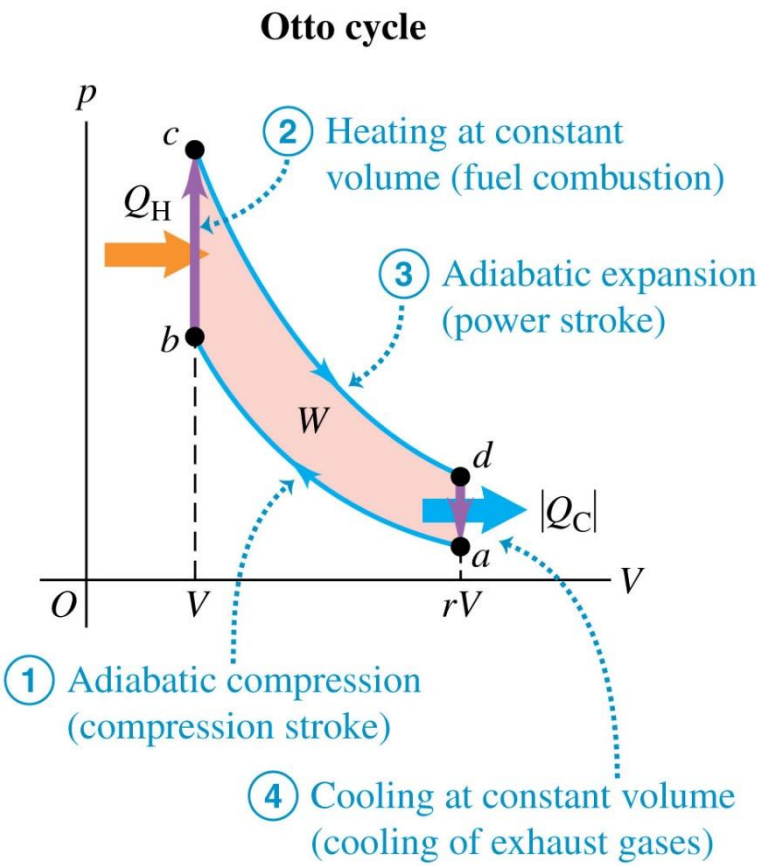
(3) **Ignition:** Spark plug ignites mixture.

(4) **Power stroke:** Hot burned mixture expands, pushing piston down.

(5) **Exhaust stroke:** Exhaust valve opens; piston moves up, expelling exhaust and leaving cylinder ready for next intake stroke.

pV-Diagram of the Otto Cycle

- Along *ab* the gasoline–air mixture is compressed adiabatically and is then ignited.
- Heat Q_H is added to the system by the burning gasoline along line *bc*, and the power stroke is the adiabatic expansion to *d*.
- The gas is cooled to the temperature of the outside air along line *da*; during this process, heat $|Q_c|$ is rejected.



Question 2

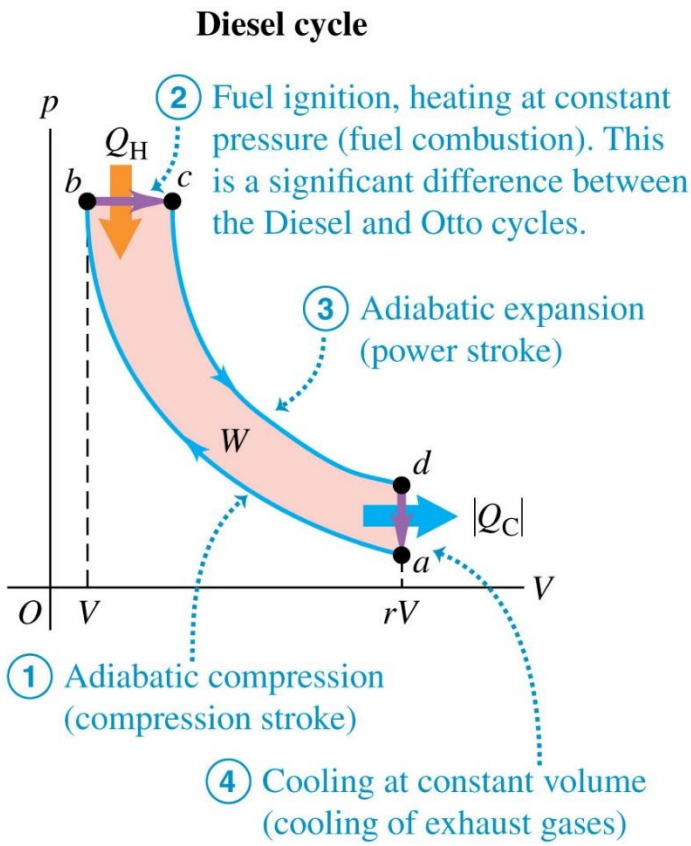
10 pts

A gasoline engine takes in 6.47 kJ of heat energy and has a thermal efficiency of 0.55. Calculate the value of Q_C (in kJ). Be careful of the sign.

-2.9115

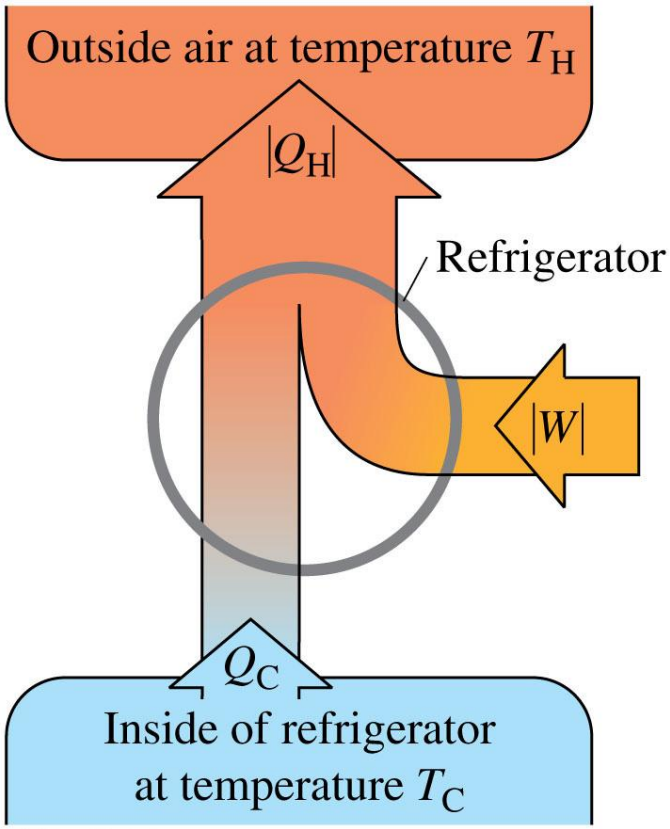
pV-Diagram of the Diesel cycle

- Starting at point *a*, air is compressed adiabatically to point *b*, heated at constant pressure to point *c*, expanded adiabatically to point *d*, and cooled at constant volume to point *a*.
- Because there is no fuel in the cylinder during the compression stroke, pre-ignition cannot occur, and the compression ratio *r* can be much higher than for a gasoline engine.
- This improves efficiency.



Refrigerators

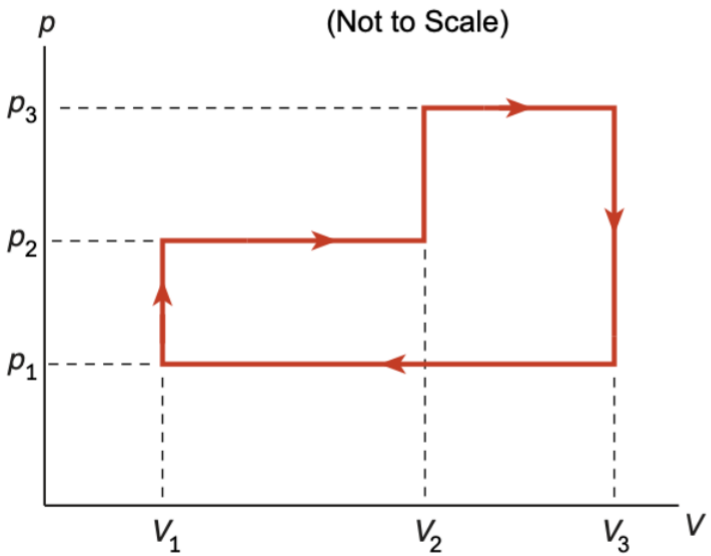
- A **refrigerator** takes heat from a cold place (inside the refrigerator) and gives it off to a warmer place (the room). An **input** of mechanical work is required to do this.
- A refrigerator is essentially a heat engine operating in reverse.
- Shown is an energy-flow diagram of a refrigerator.



Question 3

10 pts

The figure shows an engine cycle's pV-diagram, where p_1 = 93.2 kPa, p_2 = 196.3 kPa, p_3 = 289.3 kPa; V_1 = 1.1 m^3, V_2 = 2.6 m^3, and V_3 = 4.2 m^3. Calculate the net work in kJ done by the engine during one cycle.



468.41

Refrigerators: Coefficient of Performance

- From an economic point of view, the best refrigeration cycle is one that removes the greatest amount of heat from the inside of the refrigerator for the least expenditure of mechanical work.
- The relevant ratio is therefore $\frac{|Q_C|}{|W|}$ the larger this ratio, the better the refrigerator.
- We call this ratio the coefficient of performance, *K*:

Coefficient of performance of a refrigerator

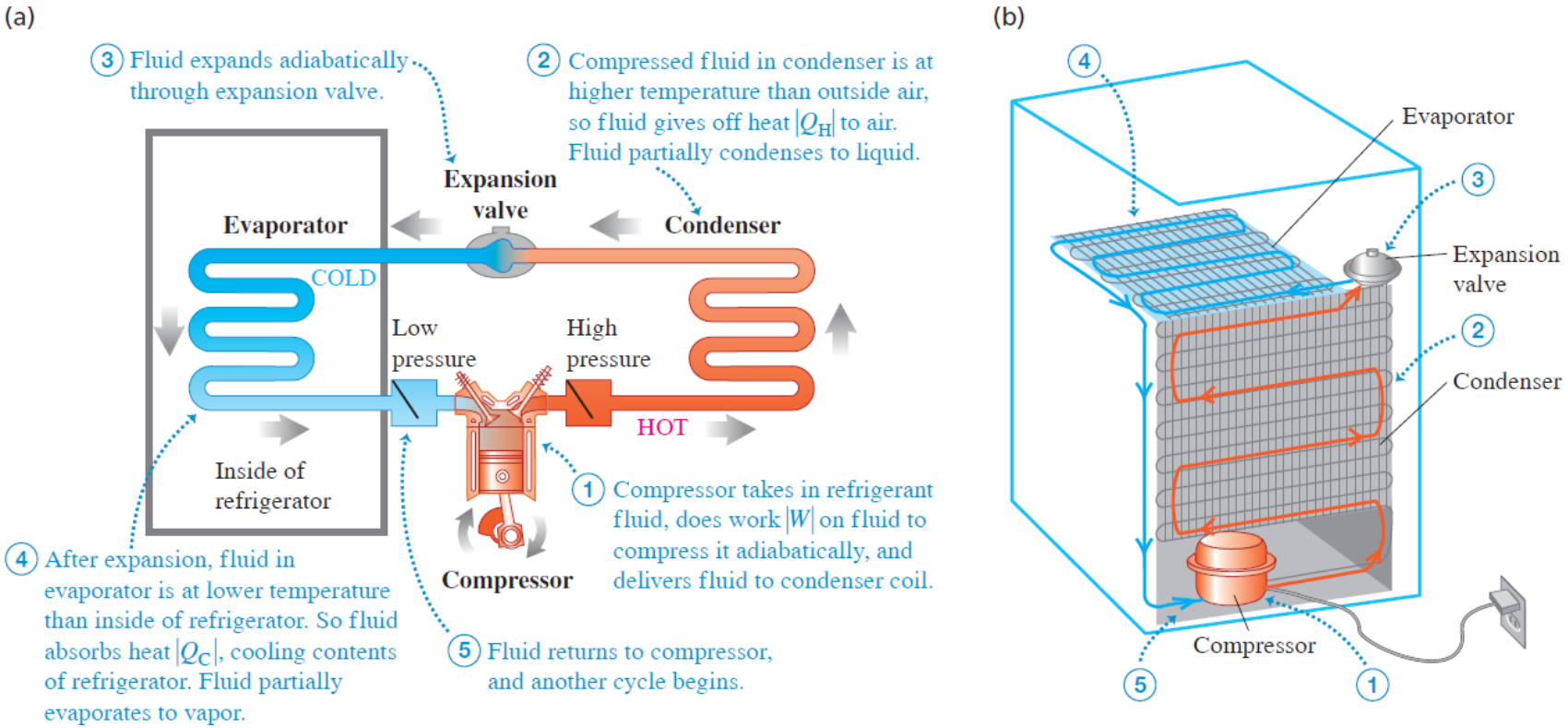
$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Heat removed from inside of refrigerator

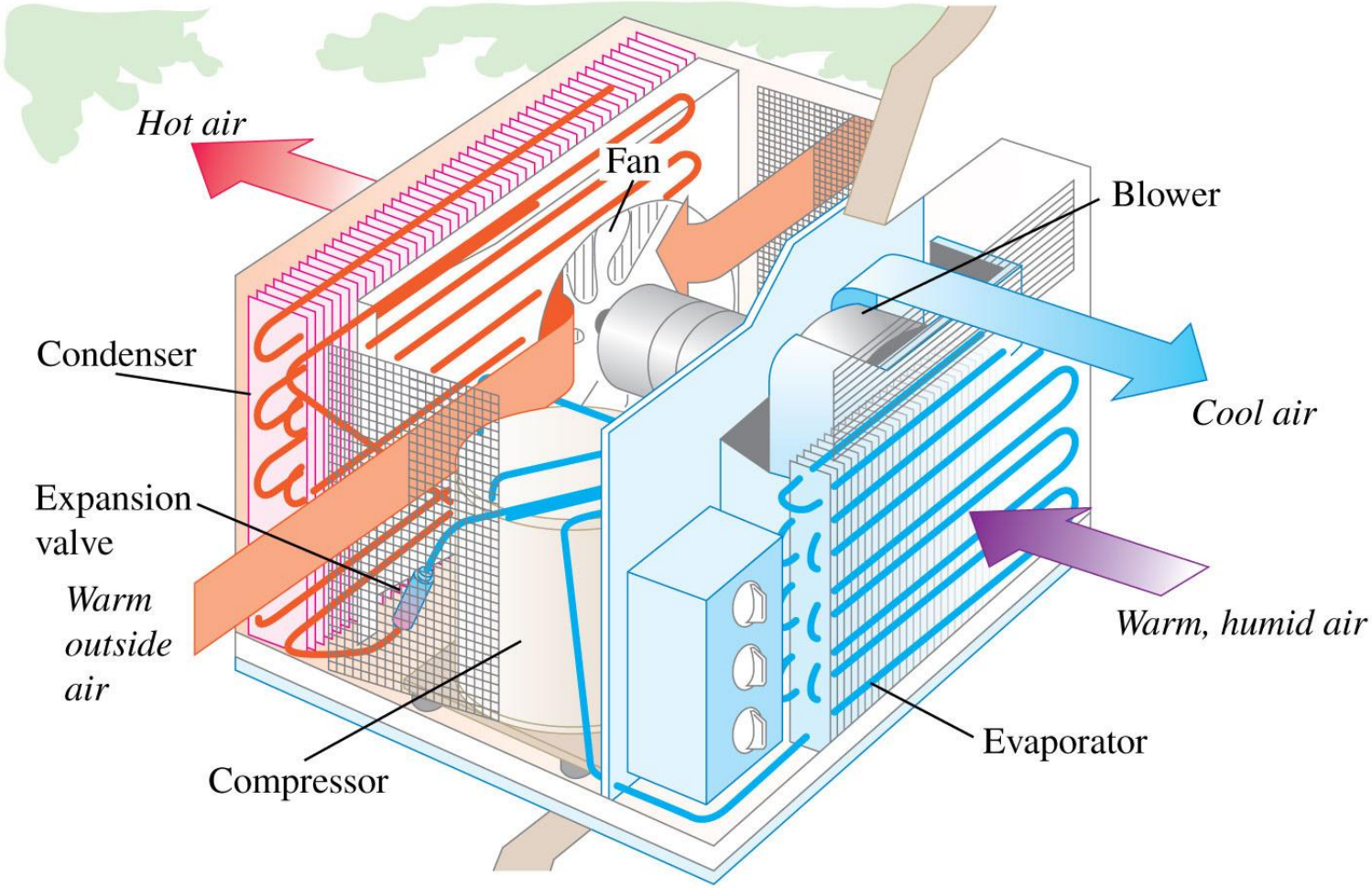
Work input of refrigerator

Heat rejected to outside air

Principle of the Mechanical Refrigeration Cycle



Air Conditioner



Question 4

10 pts

A refrigerator transfers a total of 1,748 J of energy into the kitchen while doing 231 J of work. Calculate the coefficient of performance.

6.5671

Question 5

10 pts

A heat pump in heating mode extracts 1,051 J from the cold reservoir while transferring 1,416 J to the hot reservoir. Calculate its coefficient of performance.

3.8795

Question 6

10 pts

Suppose a heat pump in cooling mode has a coefficient of performance of 9.9 and ejects 2,259 J of energy outdoors. Calculate the energy extracted from the cold reservoir.

2,052

Question 7

10 pts

An engine consists of a constant volume process at 0.8 m^3 from 131 kPa to 351 kPa, an isothermal expansion until the pressure drops to its original value, and a constant pressure process returning to its original pressure and volume. Calculate the efficiency of this process if the gas is ideal and diatomic.

[Hints: 1. You need the volume at the end of the isothermal process. Find it easily with the ideal gas law! 2. For $|Q_H|$, add the energy inputs during the constant volume and iosthermal processes.]

0.1406

The Second Law of Thermodynamics (1 of 3)

- The **second law of thermodynamics** can be stated in several ways:

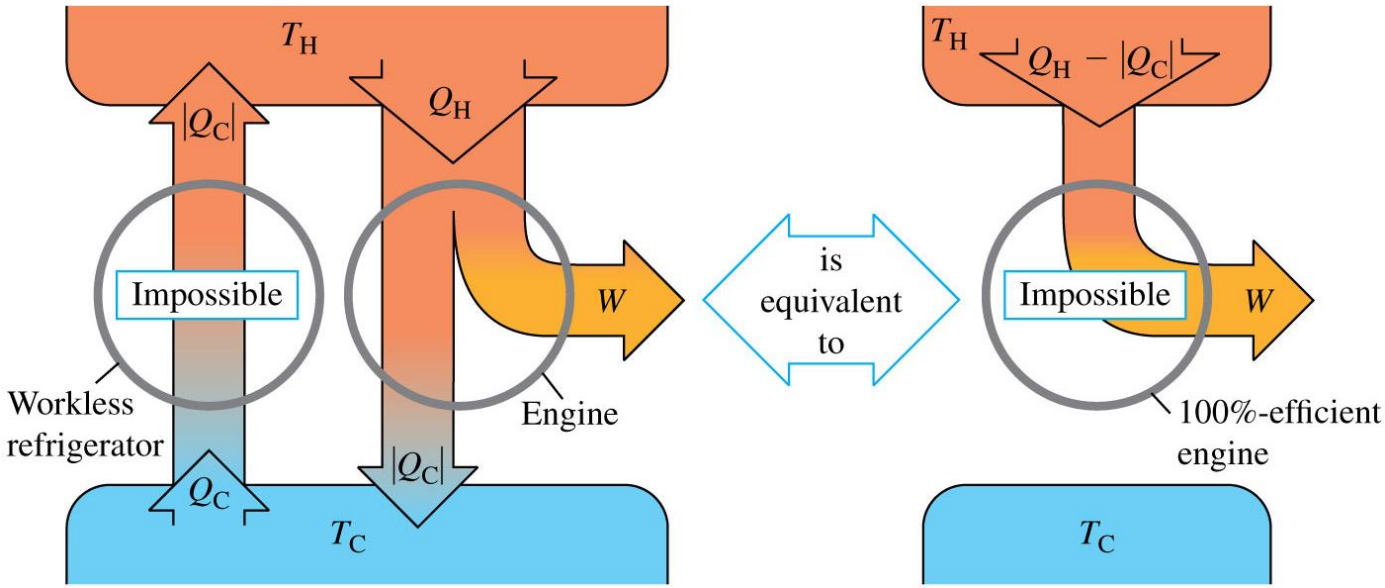
✓ **It is impossible for any system to undergo a process in which it absorbs heat from a reservoir at a single temperature and converts the heat completely into mechanical work, with the system ending in the same state in which it began.**

- We will call this the “engine” statement of the second law.

✓ **It is impossible for any process to have as its sole result the transfer of heat from a cooler to a hotter object.**

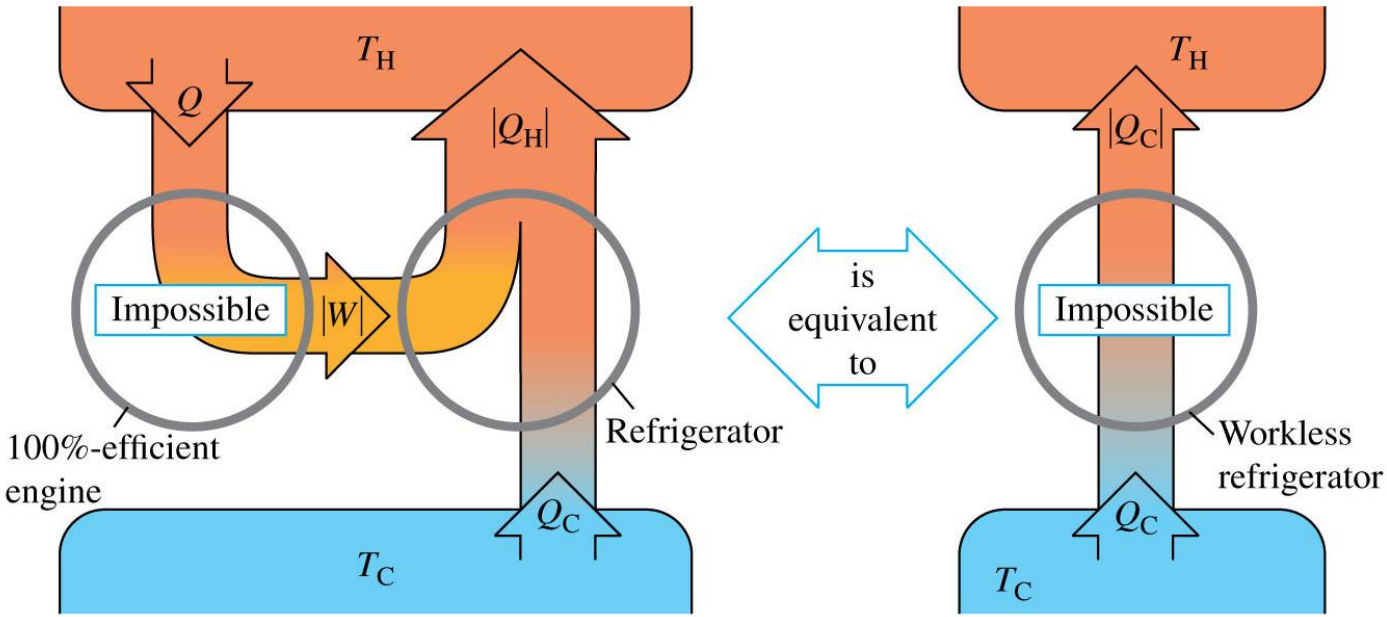
- We’ll call this the “refrigerator” statement of the second law.

The Second Law of Thermodynamics (2 of 3)



- If a workless refrigerator were possible, it could be used in conjunction with an ordinary heat engine to form a 100%-efficient engine, converting heat $Q_H - |Q_C|$ completely to work.

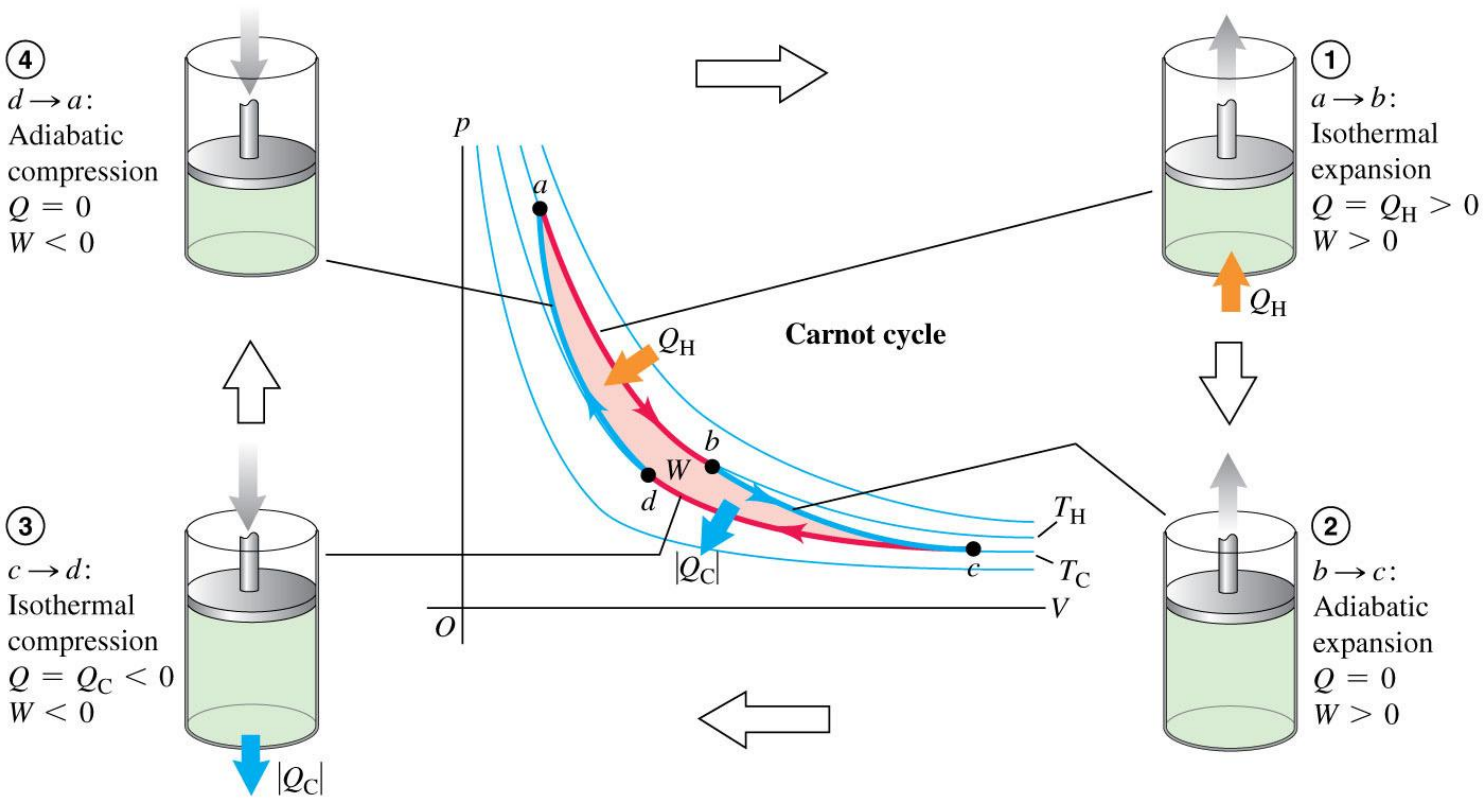
The Second Law of Thermodynamics (3 of 3)



- If a 100%-efficient engine were possible, it could be used in conjunction with an ordinary refrigerator to form a workless refrigerator, transferring heat Q_C from the cold to the hot reservoir with no input of work.

The Carnot Cycle

- A **Carnot cycle** has two adiabatic segments and two isothermal segments.



The Carnot Engine

- The Carnot cycle consists of the following steps:
  1. The gas expands isothermally at temperature T_H, absorbing heat Q_H.
  2. It expands adiabatically until its temperature drops to T_C.
  3. It is compressed isothermally at T_C, rejecting heat $|Q_c|$.
  4. It is compressed adiabatically back to its initial state at temperature T_H.
- The efficiency of a Carnot engine is:

Efficiency of a Carnot engine

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$$

Temperature of cold reservoir

Temperature of hot reservoir

- Video Tutor Solution: Example 20.2

Question 8

10 pts

An ideal engine has an efficiency of 0.56. If the temperature of the cold reservoir is 75°C, calculate the temperature of the hot reservoir in Celsius.

517.9091

Question 9

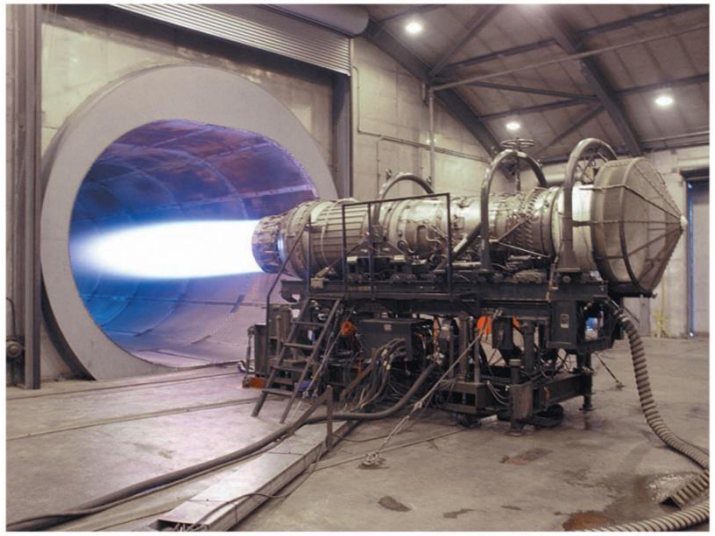
10 pts

A Carnot engine absorbs 1,240 J from the hot reservoir while doing 736 J of work. If the cold reservoir has a temperature of 21°C, calculate the temperature of the hot reservoir in Kelvin.

723.3333

Engine Efficiency

- To maximize the efficiency of a real engine, the designer must make the intake temperature T_H as high as possible and the exhaust temperature T_C as low as possible.



- For this reason, the temperatures inside a jet engine are made as high as possible.
- Exotic ceramic materials are used that can withstand temperatures in excess of 1000 degrees Celsius without melting or becoming soft.

The Carnot Refrigerator

- Because each step in the Carnot cycle is reversible, the entire cycle may be reversed, converting the engine into a refrigerator.
- The coefficient of performance of the Carnot refrigerator is:

Coefficient of performance of a Carnot refrigerator

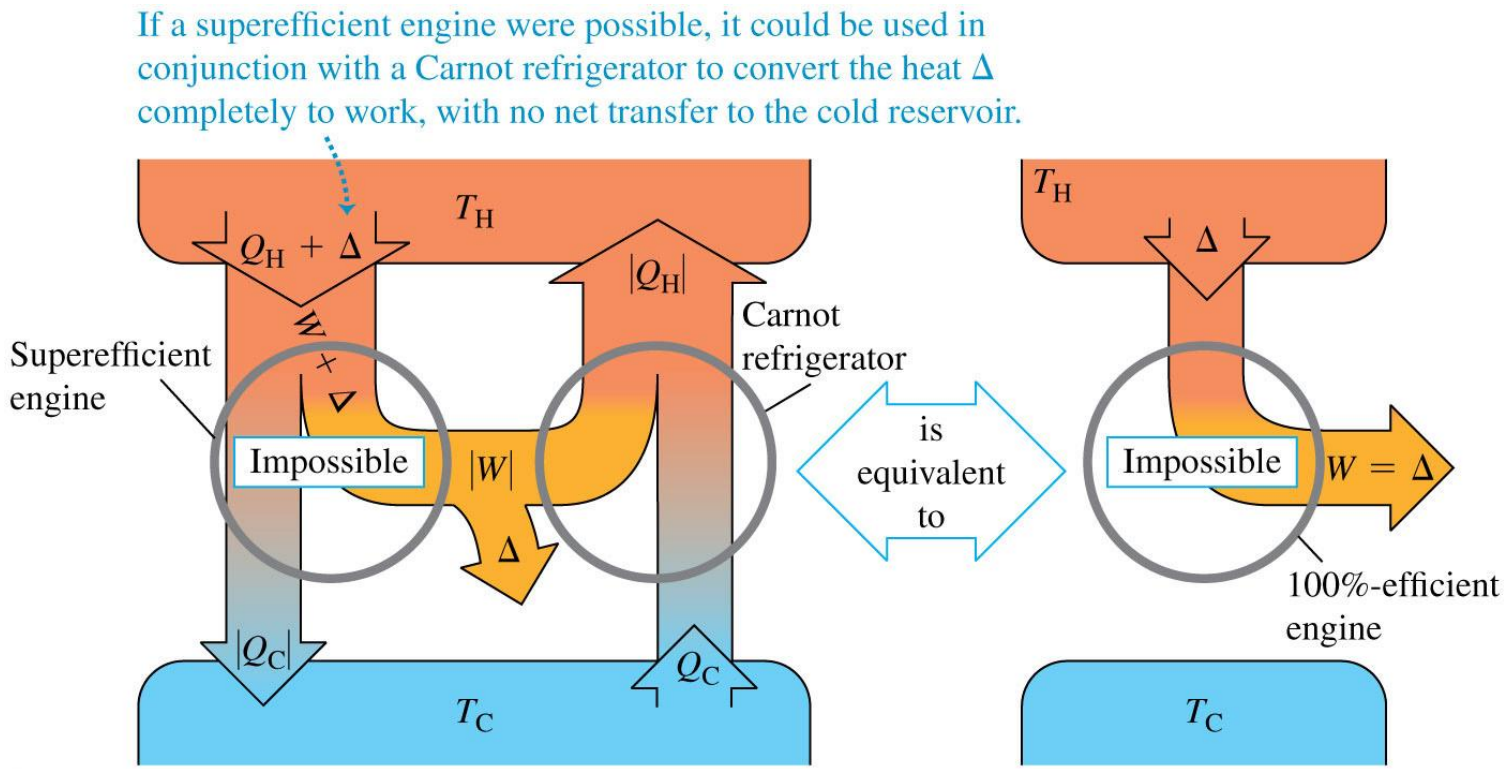
$$K_{\text{Carnot}} = \frac{T_C}{T_H - T_C}$$

Temperature of cold reservoir

Temperature of hot reservoir

The Carnot Cycle and the Second Law

- No engine can be more efficient than a Carnot engine operating between the same two temperatures.



Entropy and Disorder

- **Entropy** provides a **quantitative** measure of disorder.
- Many processes proceed naturally in the direction of increasing randomness.
- Adding heat to an object increases average molecular speeds; therefore, molecular motion becomes more random.
- The explosion of the firecracker shown increases its disorder and entropy.

Entropy in Reversible Processes

- We introduce the symbol S for the entropy of the system, and we define the infinitesimal entropy change dS during an infinitesimal reversible process at absolute temperature T as:

$$dS = \frac{dQ}{T} \quad \text{(infinitesimal reversible process)}$$

- The total entropy change over any reversible process is:

Entropy change in a reversible process

$$\Delta S = \int_1^2 \frac{dQ}{T}$$

Upper limit = final state

Infinitesimal heat flow into system

Lower limit = initial state

Absolute temperature

- Video Tutor Solution: Example 20.10

Question 10

10 pts

Suppose 7.1 kg of a solid substance is at its melting point of -33°C. If the latent heat of fusion is 36,883 J/kg, calculate the change of the substance's entropy when it melts.

ΔS = _____ J/K

1,091.1221

Question 11

10 pts

Suppose 4.5 kg of ice at 0°C melts to water at 0°C in a room that is at 15°C. The room does not significantly change temperature. Calculate the change in entropy of the universe during this process. (Latent heat of fusion of water: 3.35×10^5 J/kg.)

287.603

Question 12

10 pts

Calculate the change in entropy of water when 2.1 kg of water at 30°C warms to 57°C. [c_{water} = 4186 J/(kg·K)]

750.3643

Question 13

10 pts

An ideal monatomic gas occupying 2.3 m^3 is allowed to freely expand into an additional, evacuated volume of 2.7 m^3. Calculate the change in entropy, if there are 50.6 g of Helium gas, which has molar atomic mass of 4.00 g/mol.

81.6741

Question 14

10 pts

Suppose 11.7 kg of steam at 100°C cools to ice at 0°C. Calculate the change in entropy of the water in kJ/K.

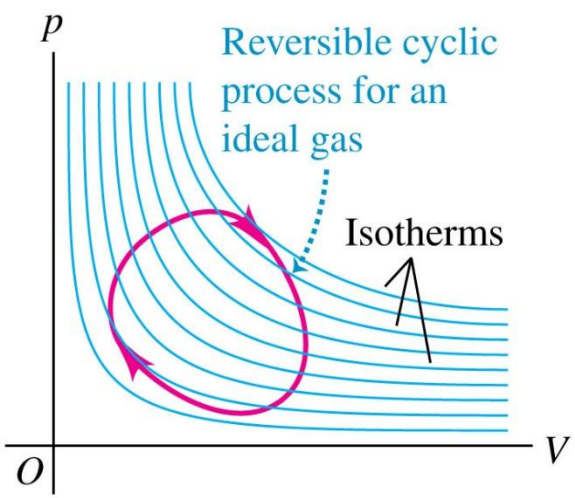
[$L_{v,water}$ = 2.26 × 10^6 J/kg, $L_{f,water}$ = 3.35 × 10^5 J/kg, c_{water} = 4186 J/(kg·K)]

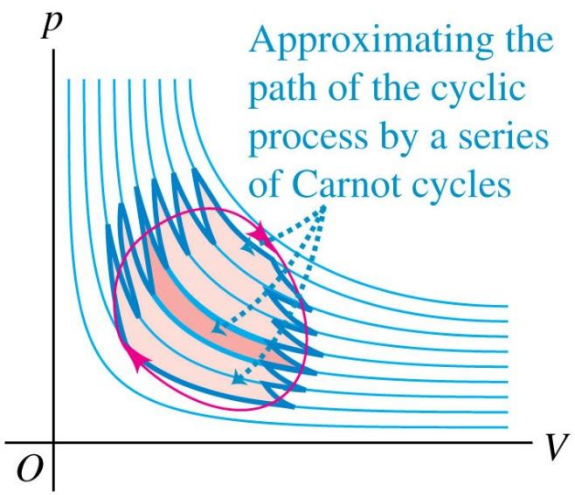
ΔS = _____kJ/K

-100.533

Entropy in Cyclic Processes

- The total entropy change in one cycle of any Carnot engine is zero.
- This result can be generalized to show that the total entropy change during **any** reversible cyclic process is zero.





Entropy and the Second Law

- The second law of thermodynamics can be stated in terms of entropy:

✓ **No process is possible in which the total entropy decreases, when all systems taking part in the process are included.**

- The entropy of the ink–water system **increases** as the ink mixes with the water.

Microscopic Interpretation of Entropy

- Suppose you toss *N* identical coins on the floor.
- The most probable outcome of tossing *N* coins is that half are heads and half are tails.
- The reason is that this macroscopic state has the greatest number of corresponding microscopic states.
- Let *w* represent the number of possible microscopic states for a given macroscopic state.
- The entropy *S* of a macroscopic state can be shown to be given by:

Expression for entropy in microscopic terms

$$S = k \ln w$$

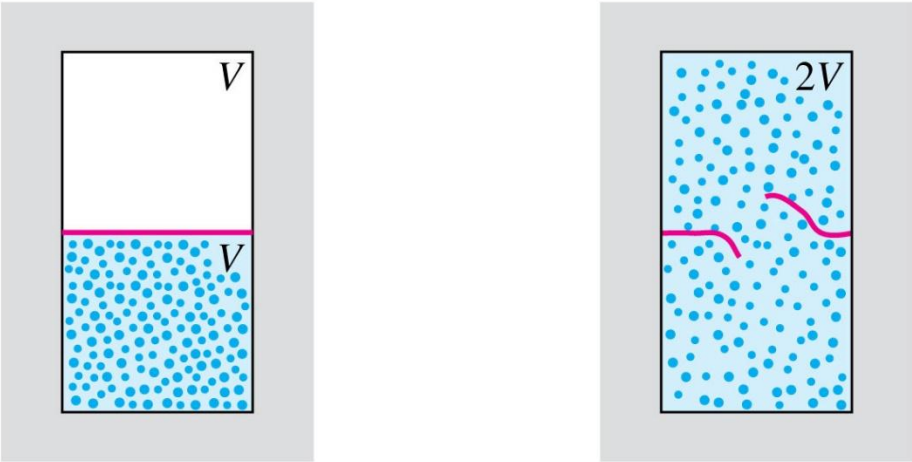
Boltzmann constant (gas constant per molecule)

Number of microscopic states for the given macroscopic state

- Video Tutor Solution: Example 20.11

A Microscopic Calculation of Entropy Change

- In a free expansion of *N* molecules in which the volume doubles, the number of possible microscopic states increases by a factor of 2*N*.

(a) Gas occupies volume V; number of microscopic states $= w_1$.

(b) Gas occupies volume $2V$; number of microscopic states $= w_2 = 2^N w_1$.